Math 61 HW 2 due 1/26

0.1 Suppose $a|b$ and $a|c$. Then, there are $d, e \in \mathbb{Z}$ such that $ad = b$ and $ae = c$. After substituting b and c, $a|(b+c) = a|(ad+ae) = a|a(d+e)$. Since the right side has a as a factor, a divides $a(d+e)$ and therefore $b+c$ as well. $a|(b+c) = a|a(d+e)$.

0.2 Prove that for all integers m and n, if m is odd and n is even, then mn is even.

(For every $m, n \in \mathbb{Z}$, m odd n even, mn even)

Proof: Suppose m is odd and n is even. By definition of odd, $m = 2k+1$ for some $k \in \mathbb{Z}$. By definition of even, $n = 2\ell$ for some $\ell \in \mathbb{Z}$. $mn = (2k+1)(2\ell) = 2k\ell + 2\ell$. This of the form "$2x$", $2(k\ell + \ell)$, where $x = k\ell + \ell$. Since mn is divisible by 2, mn is even.

0.3 Let x and y be real numbers. Prove that if $x \leq y + \varepsilon$ for every positive real number ε, then $x \leq y$.

Proof: Suppose $x \leq y + \varepsilon, \forall \varepsilon > 0$. For the sake of contradiction, let $x > y$. There exists a distance between x and y, h. $x - y = h \Rightarrow x = y + h$. Let $\varepsilon < h$. $y + h > y + \varepsilon$, so we could conclude $x \leq y + \varepsilon < y + h$. But x must equal $y + h$ when $x > y$, so this conclusion is nonsense / false. Thus $x \leq y$.

0.4 Converse: If x is irrational, then x^2 is irrational.
Let $x = \sqrt{2}$, which is irrational. $x^2 = (\sqrt{2})^2 = 2$. 2 is rational. (by definition $2 = \frac{A}{B}$) ✗
The converse of ex 1 is false.

0.5 The contrapositive is for every $x \in \mathbb{Z}$, if x is even, then $x^3 - 1$ is odd. If x is even, $\exists k \in \mathbb{Z}$ such that $x = 2k$. Plugging this in to $x^3 - 1$, $(2k)^3 - 1 = 8k^3 - 1$. By definition of odd, we can rewrite this equation as $2(4k^3) - 1$. $\exists l \in \mathbb{Z}$ such that $l = 4k^3$. This means $x^3 - 1$ is indeed odd. Because the contrapositive is true, $P \Rightarrow Q$.

0.6 Suppose that for every $a, b \in \mathbb{R}$, a is rational and ab is irrational. For the sake of contradiction, b is rational. A real number $x \in \mathbb{R}$ is called rational if $x = \frac{A}{B}$ for some integers $A, B \in \mathbb{Z}$. $B \neq 0$. a is rational so $a = \frac{A}{B}$ and b is rational so $b = \frac{C}{D}$. $a * b = \frac{A}{B} * \frac{C}{D} = \frac{AC}{BD}$. $A \cdot C = E$ and $BD = F$ for some $E, F \in \mathbb{Z}$. $F \neq 0$. $ab = \frac{E}{F}$, which is rational by definition. This contradicts the statement that ab is irrational. Thus, if a is rational and ab is irrational, then b must be irrational.

0.7 TP $a, b \in \mathbb{R}_{\geq 0}$, then $a + b \geq 2\sqrt{ab}$. After squaring both sides, $(a+b)^2 \geq 4ab$.
$(a+b)^2 - 4ab \geq 0$.
$a^2 + 2ab + b^2 \geq 0$
$(a-b)^2 \geq 0$
The square of any real number is always positive, so the inequality is true for all $a, b \in \mathbb{R}_{\geq 0}$. Thus if $a, b \in \mathbb{R}_{\geq 0}$, then $a + b \geq 2\sqrt{ab}$.